LESSON 2.9
FLUENCY WITH OPERATIONS

LESSON INTRODUCTION

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.

LEARNING TARGETS

- I can fluently multiply and divide multi-digit numbers with decimals to the thousandths.
- I can fluently multiply and divide positive fractions by positive fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.2.2 Extend previous understanding of multiplication to multiply a fraction by a fraction, including mixed numbers and fractions greater than 1, with procedural reliability.

MA.5.NSO.2.5 Multiply and divide a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

ESSENTIAL QUESTIONS

- How can I use a growth mindset to improve in accuracy and fluency with decimal and fraction operations?
- Which methods of multiplying and dividing decimals best contribute to fluency?

LESSON NARRATIVE

This lesson is focused on fluency of operations with decimals and fractions. After a brief Guided Practice where students can work with partners to practice what they have learned in prior lessons, the lesson moves to a game and additional practice to build procedural fluency and accuracy.

COMPONENT SUMMARY

2.9.1 WARM-UP *(~5 MIN.)*

Math anxiety and growth-mindset discussion

2.9.2 GUIDED PRACTICE *(10 MIN.)*

Partner work reviewing multiplying and dividing decimals and fractions

2.9.3 ACTIVITY *(15-20 MIN.)*

Partners play “Fraction War” card game

2.9.4 YOUR TURN! *(10-15 MIN.)*

Students practice multiplying and dividing decimals and fractions

2.9.5 WRAP-UP *(~5 MIN.)*

“Confidence Scale” self-assessment on learning targets from lesson and unit

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Deck of cards (one per pair)
- (Optional) Individual mini whiteboards
- (Optional) Whiteboard markers

ADDITIONAL PREPARATION

- If using some class time for small-group additional practice, consider reviewing formative assessments from prior lessons to create small groups before the lesson begins.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse algorithms for multiplying and dividing decimals and fractions.
- Students may misinterpret products or quotients with tenths, hundredths, and thousandths.

THINGS TO CONSIDER

- This lesson uses skills from Unit 2 on multiplying and dividing fractions and decimals. Consider using 2.9.2 Guided Practice as an opportunity for students to recall the various strategies and procedures.
- Consider modeling how to check work and identify computational errors through 2.9.2 Guided Practice. This may help foster a growth mindset and build confidence, as minor mistakes often occur when computing by hand.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

For the problems in 2.9.4 Your Turn!, consider using individual mini whiteboards or other ways for students to show individual responses during whole group practice. Have students individually solve each problem on the whiteboard, then show them to group members or hold them up for the teacher to see in order to quickly assess learning and highlight specific student responses for further discussion.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

This lesson focuses on strengthening procedural fluency for operations with decimals. Clarifications for this standard include providing multiple opportunities for students to practice methods. Consider helping students maintain accuracy while performing algorithm or modeling strategies, and helping students learn from any computational mistakes to build confidence.

DIFFERENTIATE INSTRUCTION

Consider splitting the class after 2.9.2 Guided Practice, having some students play the game while others work on 2.9.4 Your Turn! problems, and vice versa. During this time, the teacher could pull small groups for additional support or small-group instruction based on formative assessments from the prior lessons.

2.9.3 Activity Fraction War – all learning styles can be present and active during this interactive activity. Make space for students to lean into their strengths and styles while demonstrating proficiency in this benchmark.

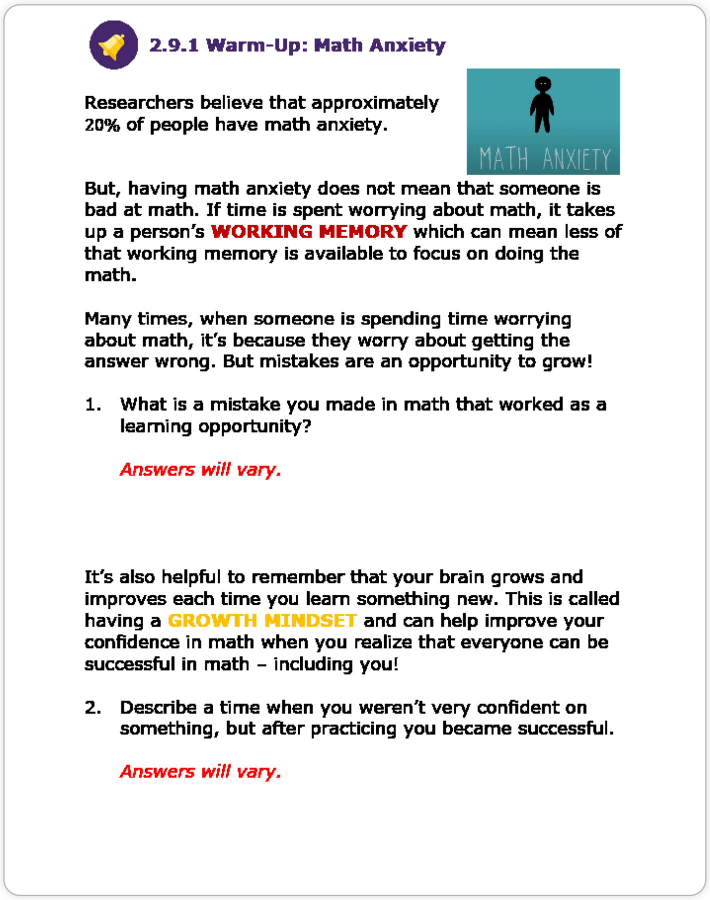
Consider what formative and summative assessments look like as you close the unit. Consider digital practice tools like Check Your Understanding, EdgeXL, and Test Yourself! In addition to the focus of growth mindset (Warm-Up) and qualitative feedback (surveys or Wrap-Up) when assessing students.

2.9.1 WARM-UP: MATH ANXIETY

Component Overview: *Students recall a previous video on math anxiety and how it affects working memory and a growth mindset. Use the two open-ended Warm-Up reflection questions to have students share responses in partners or as a whole group, then write down responses.*

TEACHER MOVES

Consider having students share their ideas in partners or as a whole group rather than writing their answers. Also, consider modeling this reflection by using a think-aloud as an example for students first. Use sentence stems to encourage reflection such as, "A mistake I made in math that I learned from was…" or "Something I wasn't very confident in at first was…"


2.9.1 Warm-Up: Math Anxiety

Researchers believe that approximately 20% of people have math anxiety.

But, having math anxiety does not mean that someone is bad at math. If time is spent worrying about math, it takes up a person's **WORKING MEMORY** which can mean less of that working memory is available to focus on doing the math.

Many times, when someone is spending time worrying about math, it's because they worry about getting the answer wrong. But mistakes are an opportunity to grow!

1. What is a mistake you made in math that worked as a learning opportunity?

   Answers will vary.

It's also helpful to remember that your brain grows and improves each time you learn something new. This is called having a **GROWTH MINDSET** and can help improve your confidence in math when you realize that everyone can be successful in math – including you!

2. Describe a time when you weren't very confident on something, but after practicing you became successful.

   Answers will vary.

2.9.2 Guided Practice: Fluency with Operations

1. Work with your partner to find each product or quotient. Share strategies and discuss any questions you still have about how to perform each operation.

a. 15.5×4.87

75.485

b. $3\frac{1}{5} \div \frac{7}{4}$

$\frac{16}{5} \cdot \frac{4}{7}$

$= \frac{64}{35}$ *or* $1\frac{29}{35}$

c. $\left(2\frac{4}{9}\right)\left(6\frac{7}{4}\right)$

$\frac{22}{9} \cdot \frac{31}{4}$

$= \frac{682}{36}$

$= \frac{341}{18}$ *or* $18\frac{17}{18}$

d. $1.162 \div 8.3$

0.14

2.9.2 GUIDED PRACTICE: FLUENCY WITH OPERATIONS

Component Overview: *Students work in partners to recall strategies and discuss questions when finding products and quotients. This section can serve to summarize operations with decimals and fractions. If some students are having difficulty remembering procedures, consider having them review past work in this unit to recall strategies.*

TEACHER MOVES

Throughout this practice section, encourage students to use correct math vocabulary when identifying parts of expressions or steps in procedures and sharing strategies. Ensure students are using operation-specific vocabulary correctly, including product, factor, divisor, dividend, and quotient.

ENRICHMENT

Consider asking students:

- What strategy did you use to find the product/quotient?
- Are there any other ways you could have solved for the product/quotient?
- How can you check if the product/quotient is correct?

2.9.3 ACTIVITY: FRACTION WAR

Component Overview: *Students use a card deck to play Fraction War in partners. Review the setup, rules, and "how to play" section as a class and model playing with a student, exemplifying how to compute the answer using a think-aloud and specific math vocabulary. This activity may also be adapted so half of the class plays while the other half completes Your Turn! problems. Use ~10-minute rotations to allow students to see a variety of fraction multiplication and division problems.*

DIFFERENTIATED INSTRUCTION

Fraction War activates a variety of strengths and learning styles for students.

- Kinesthetic Entry: Students have the ability to construct and manipulate concrete representations in real time. Additionally, students can rotate with partners based on whose product or quotient is higher in value.
- Visual Entry: Students practice navigating the concrete models with their abstract concepts of fractions and operations.
- Auditory Entry: Students can sit back-to-back and be required to describe or explain their cards, their fractions, and their product or quotient.

TEACHER MOVES

If you are looking for a variation for an engaging activity later in the unit, consider incorporating decimals into the game. Rather than cards representing fractions, each card could represent different place values for decimals for students to practice.

QUESTIONING

Consider having students use an area model or tape diagram for one round, then switch to using the standard algorithm for the next. Using more than one strategy will increase procedural fluency and conceptual understanding of fraction operations.

2.9.3 Activity: Fraction War

Setup:

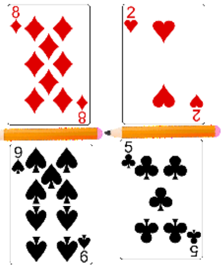
- With your partner, separate the playing cards into two stacks. The first stack will contain only black cards (Spades ♠ and Clubs ♣). The other stack will contain only red cards (Diamonds ♦ and Hearts ♥). Any jokers should be removed from the deck.
- Place two pencils on the desk or table to be the fraction bars in the game.

Rules:

- The black cards (Spades ♠ and Clubs ♣) are denominators.
- The red cards (Diamonds ♦ and Hearts ♥) are numerators.
- Aces = 1; Jacks = 11; Queens = 12; and Kings = 13.

How to Play:

To begin, each player will flip over a card from each stack to create two fractions, as shown.



- Each player performs the operation indicated for the round. Correct answers earn 1 point and incorrect answers earn 0 points. Both players can earn points in any round.
- Each round, the operation switches between multiplication (×) and division (÷).
- When a player gives an answer, both players will check the answer provided. A correct answer is the only way to move to the next round.
- Repeat this process until time is called.

Scoreboard		**Workspace**
Player A:	**Player B:**	

2.9.4 Your Turn!

1. Find each product or quotient. Show your work.

a. $26.22 \div 9.6$
2.73125

b. 15.928×1.6
25.4848

c. $81.2 \div 2.9$
28

d. $\frac{7}{8} \times \frac{7}{21}$
$\frac{49}{168}$
$= \frac{7}{24}$

e. $3.834 \div 1.42$
2.7

f. 7.23×9.4
67.962

g. $2\frac{1}{51} \cdot 1\frac{1}{3}$
$\frac{103}{51} \cdot \frac{4}{3}$
$= \frac{412}{153}$ or $2\frac{106}{153}$

h. $(21.8)(15.12)$
329.616

i. $3\frac{5}{9} \div \frac{2}{3}$
$\frac{32}{9} \div \frac{2}{3}$
$= \frac{32}{9} \cdot \frac{3}{2}$
$= \frac{96}{18}$
$= \frac{16}{3}$ or $5\frac{1}{3}$

j. 11.516×2.3
26.4868

k. $1\frac{6}{7} \div 5\frac{3}{4}$
$\frac{13}{7} \div \frac{23}{4}$
$= \frac{13}{7} \cdot \frac{4}{23}$
$= \frac{52}{161}$

2.9.4 YOUR TURN!

Component Overview: *Students will practice finding the product or quotient to a variety of problems. This activity may be adapted to happen concurrently with the previous 2.9.3 Activity Fraction War as rotations or can be used as an opportunity to engage in whole-group practice using Poll the Class routines. Consider your students' learning styles and interest in engaging in various forms of procedural practice with accuracy.*

TEACHER MOVES

Poll the Class

Consider using this section to engage all students in Poll the Class routines. Invite students to individually solve each problem on a mini whiteboard, then show their work and responses to the teacher. This practice can be used as a formative assessment for the teacher to quickly determine procedural understanding and highlight specific student responses by having that student share their thinking with the class. Also consider highlighting how students can learn from computational mistakes.

QUESTIONING

For any students who finish early with one strategy, consider giving a challenge strategy that students can try after they finish with their originally chosen strategy. This challenge could be using the distributive property or a diagram or model.

2.9.5 WRAP-UP: REFLECTION

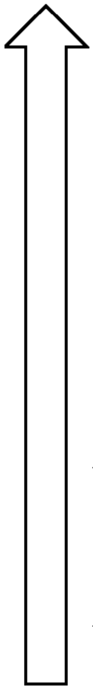
Component Overview: *This open-ended Wrap-Up is a self-reflection for students to assess their learning and confidence in operations with decimals and fractions from this unit. Use this self-assessment of learning to help students reflect on effort and a plan for getting help if needed. Consider checking in with students individually on their plan for getting help after the lesson.*

DIFFERENTIATED INSTRUCTION

Coloring in the arrow could serve as an entry point for visual learners as they reflect on their learning this unit. Consider planning short individual check-ins as part of small-group remediation as you conclude the unit and plan for end-of-unit assessment.

2.9.5 Wrap-Up: Reflection

Color in the arrow up to the statement that best describes your current level of confidence with multiplying and dividing fractions and decimals.
Answers will vary.



I'm very confident! I can perform each operation and I could explain this to someone else.

I can get to the right answer when performing each operation, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. Check off what you do understand.
- ☐ Multiplying decimals
- ☐ Dividing decimals
- ☐ Multiplying fractions
- ☐ Dividing fractions

I have been trying hard and listening, but I am finding this challenging. I will get help by ______ ____.

I do not understand any of this yet. I will get help by ________________________________ ____.